



CARIVIX – *Ask. Visualize. Achieve.*
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CARIVIX AI

Weekly Team Progress Report

Field	Details
Project	CARIVIX AI – AI-Powered Research, Intelligence & Decision Support Ecosystem
Report Type	Weekly Team Progress Report
Sprint	Sprint 1 – Foundation & Project Setup
Week	Week 1
Reporting Period	20/07/2026 – 25/07/2026
Report Date	25/07/2026
Overall Project Status	● On Track

1. Executive Summary

During **Sprint 1 (Foundation & Project Setup)**, the CARIVIX AI team made significant progress in establishing the project's technical, analytical, and documentation foundations. The team successfully completed activities across Python development, Artificial Intelligence, Machine Learning, GIS Engineering, Data Science, NLP, Data Analytics, Technical Writing, Research, SEO/SEM, UI/UX Design, Branding, and Graphic Design. Major accomplishments included developing reusable Python helper modules, preparing and validating datasets, configuring GIS services, researching multilingual AI technologies, executing machine learning models, creating analytical dashboards and visualizations, conducting competitive AI market research, preparing API and developer documentation, and defining AI development strategies. These deliverables provide a strong and scalable foundation for the next phase of CARIVIX AI development.

The sprint also focused heavily on enterprise readiness through structured documentation, coding standards, KPI planning, developer guides, content strategies, and business intelligence research. Teams successfully completed market gap analysis, technology comparisons, AI roadmap planning, predictive analytics research, GIS infrastructure preparation, SEO strategy documentation, onboarding UI designs, and brand asset development.

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Overall, more than **90% of the assigned Sprint 1 tasks were successfully completed**, with only a few documentation and design-related activities remaining, including the documentation repository structure, document naming conventions, mood board preparation, and enterprise presentation standards. These remaining tasks are scheduled for completion in the following working day, keeping **Sprint 1 on track and the CARIVIX AI project well-prepared for the implementation phase**.

2. Overall Project Progress

Module	Major Deliverables	Status
Python/R Developer	- Set up database connectivity - Create a logging framework. - Install Python development environment - Configure virtual environment - Set up project repository - Set up database connectivity - Create a logging framework. - Build data preprocessing scripts.	Completed All assigned tasks
GIS Engineer	- Verify dataset formats - Organize datasets into a structured repository. - Install QGIS and GIS development tools. - Identify required GIS data sources. - Import GIS datasets into QGIS. - Validate map projections and coordinate reference systems (CRS). - Fix geometry errors and clean datasets. - Administrative Boundary Updates. - Infrastructure (Roads) and Hydrology (Water Bodies) Integration.	Inprogress: Prepare base layers (administrative boundaries, roads, water bodies). And Create attribute tables.
NLP/Voice Assistant Developer	- Researching text preprocessing techniques. - Implementing tokenization. - Testing sentence and word segmentation. - Utilizing inbuilt datasets for demonstration due to the absence of external data. - Install NLP Libraries (NLTK, spaCy, Transformers) - Configure Development Environment	Completed All assigned tasks

	• Study CARIVIX NLP Architecture • Implement stemming and lemmatization. • Compare algorithm performance. • Document preprocessing outputs.	
Data Scientist	• Research publicly available machine-learning datasets relevant to CARIVIX AI. • Classify datasets based on business domain. • Document dataset sources and licensing information. • Import selected datasets into Python. • Perform Exploratory Data Analysis (EDA). • Analyse dataset structure, features, and distributions. • Detects missing values, outliers, and inconsistencies. • Evaluate data quality and completeness. • Recommend suitable preprocessing approaches.	Completed All assigned tasks
Data Analyst	• Study dashboard requirements for executive and operational users. • Define visualization requirements. • Research economic indicators relevant to CARIVIX AI. • Organize datasets by category. • Understand CARIVIX AI reporting requirements. • Identify business metrics and KPIs for each module. • Identify government data sources. • Collect sample public datasets. • Analyze dataset quality. • Detect missing values and inconsistencies.	Completed All assigned tasks
Research Analyst	• Analyze Microsoft Copilot. • Document strengths, limitations, and enterprise features. • Study the current AI landscape. • Identify major AI products and competitors. • Study ChatGPT.	Completed All assigned tasks

	● Evaluate conversational AI capabilities, plugins, integrations, and enterprise applications. ● Analyze Google Gemini. ● Compare multimodal capabilities and AI ecosystem	
ML & AI Engineer	● Study the CARIVIX AI vision, objectives, and functional requirements. ● Identify the main business domains. ● Identify potential AI prediction use cases. ● Assist the ML & AI Engineer in data preprocessing and environment testing. ● Review sample machine learning notebooks. ● Study Python coding standards and project structure. ● Understand documentation practices. ● Install Python, Git, VS Code, and required AI libraries. ● Configure the development environment. ● Learn Git branching strategy. ● Practice clone, commit, push, pull, and merge operations.	Completed All assigned tasks
Graphic Designer	● Design 3–5 logo concepts for CARIVIX AI. ● Explore typography and color palette options. ● Present initial logo ideas for review. ● Study the CARIVIX vision, mission, and target audience. ● Research AI and enterprise software branding trends. ● Create a mood board and visual inspiration. ● Design 3–5 logo concepts for CARIVIX AI. ● Explore typography and color palette options. ● Present initial logo ideas for review. ● Design 3–5 logo concepts for CARIVIX AI. ● Explore typography and color palette options. ● Present initial logo ideas for review. ● Finalize the selected logo.	Completed All assigned tasks

	● Create logo variations (horizontal, vertical, monochrome, favicon). ● Define brand color palette and typography.	
SEO/SEM Specialist	● Develop a backlink acquisition strategy. ● Identify potential outreach websites and partners. ● Analyze competitor websites. ● Evaluate keyword rankings and content strategies. ● Research high-value keywords related to AI, analytics, GIS, NLP, and business intelligence. ● Group keywords by search intent. ● Research high-value keywords related to AI, analytics, GIS, NLP, and business intelligence. ● Group keywords by search intent. ● Draft meta titles and meta descriptions for key pages. ● Prepare SEO-friendly URL structure	In Progress Lack of access to premium SEO tools (such as SEMrush or Ahrefs) limited the ability to perform an accurate keyword ranking analysis.
Technical Writer	● Create the Documentation Repository Structure. ● Define Document Naming Conventions. ● Draft API documentation templates. ● Prepare endpoint documentation format. ● Develop the User Manual outline. ● Create administrator and user documentation structure. ● Draft the Developer Guide. ● Document Git workflow, coding practices, and contribution guidelines.	Completed All assigned tasks

3. Team Member Progress

Name	Role	Total Tasks Assigned	Tasks Completed	Status
Sindhu Bollikonda	Python/R Developer	09	09	 Completed
V. V. Naga Raju Neetipalli	GIS Engineer	15	12	In Progress
Ranjith Kumar Ashadapu	Technical Writer	13	13	 Completed
Bhanu Durgesh Dammala	Data Scientist	11	11	 Completed
Deepak Manthena	NLP/Voice Assistant Developer	15	15	 Completed
Sravan Kumar	NLP & Voice Assistant Developer	11	00	In Progress
Shaik Subhani	Data Scientist	14	14	 Completed

Nandhu Mamidi	Data Analyst	09	09	 Completed
Sri Abhinav Nagamalla	Data Analyst	12	12	 Completed
Roshan Kumar	Research Analys	10	10	 Completed
Nerella Raghavendra Guptha	ML & AI Engineer	08	08	 Completed
Srinidhi	Graphic Designer	05	05	 Completed
Akula Shiva Kumar	Graphic Designer	09	09	 Completed
Akash Seloju	SEO/SEM Specialist	10	10	 Completed
Namitha Sahaay	SEO/SEM Specialist	10	09	In Progress
Shivanath Samudrala	Technical Writer	14	14	 Completed

Manga Sai Nikhil	ML & AI Engineer, (Intern)	10	10	 Completed
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4. Project Statistics

Category	Count
Total Team Members	19
Total Tasks Assigned	193
Tasks Completed	187
Tasks In Progress	6
Tasks Pending	6
Completion Percentage	96.89 %

5. Total Task Deliverables and Status

S.No	Domain / Module	Total Deliverables	Completed	In Progress	Pending	Status
1	Python/R Development	9	9	0	0	✓ Completed
2	GIS Engineering	15	12	3	0	● In Progress
3	NLP / Voice Assistant Development	26	23	3	0	● In Progress
4	Data Science	25	25	0	0	✓ Completed
5	Data Analytics	21	21	0	0	✓ Completed
6	Research & Competitive Analysis	10	10	0	0	✓ Completed
7	ML & AI Engineering	18	18	0	0	✓ Completed
8	Graphic Design & Branding	14	14	0	0	✓ Completed
9	SEO / SEM	20	19	1	0	● In Progress

10	Technical Documentation	27	27	0	0	 Completed
—	Overall Project Deliverables	193	187	6	0	96.89% Completed

6. Key Achievements

- Verified the accuracy and completeness of the **TELANGANA_DISTRICTS** dataset, confirming all **33 administrative districts** are correctly represented.
- Resolved the DB Manager access issue by using an alternative workflow
- Validated the preprocessing pipeline using the **NLTK Gutenberg Corpus** and **Movie Reviews Corpus**, confirming consistent and accurate text preprocessing across multiple datasets.
- Measured preprocessing performance using token count reduction, token reduction percentage, and execution time metrics to evaluate efficiency and effectiveness.
- Developed a comprehensive feature selection and feature engineering strategy covering numerical, categorical, time-series, text, and location-based data. prepared comprehensive KPI documentation for both **Government Analytics** and **Economic Analytics** datasets.
- Identified and documented **115 unique KPIs** across six datasets, including Economic Trend Analysis (44), Employment Analytics (7), GDP & Inflation Forecasting (6), Infrastructure Monitoring (12), Policy Impact & Budget Analysis (21), and Public Program Evaluation (25).
- Defined six implementation phases, including the **AI Research Platform, Business Intelligence Platform, Economic Intelligence Platform, Government Intelligence Platform, Smart City Intelligence Ecosystem, and Global Intelligence Network**.
- Developed an interactive control panel for boundary layer visibility and real-time zoom monitoring, improving GIS visualization usability.
- Documented a standardized **Git workflow**, including branching strategy, Conventional Commit guidelines, pull request process, and release management to support efficient collaboration.
- Drafted a comprehensive **Developer Guide** for the CARIVIX AI platform, covering project architecture, development environment setup, workflows, release processes, and contributor onboarding.
- Executed multiple sample machine learning models, including **Linear Regression, Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN)**, to validate the machine learning environment and workflow.
- Planned **24 SEO-optimized blog topics** across 12 weeks, mapped to key buyer personas and different stages of the marketing funnel.
- Designed dedicated **landing page strategies** for Business Intelligence, Government Intelligence, Smart City Intelligence, Research Intelligence, and Pricing to improve lead generation and conversion.
- completed a comprehensive competitive landscape analysis covering leading AI platforms including ChatGPT, Gemini, Claude, Microsoft Copilot, Perplexity AI, and DeepSeek.

- developed reusable Python helper modules aligned with the CARIVIX AI system architecture to support data acquisition and data processing workflows
- Prepared comprehensive technical documentation and a README file describing the module architecture, project structure, required dependencies, installation instructions, and usage examples to support future development and integration.
- Designed a complete website information architecture and content strategy, including optimized navigation, pillar pages, blogs, whitepapers, case studies, product documentation, FAQs, and resource pages to strengthen topical authority and improve user experience.

7. Challenges / Issues

1. From NLP/ Voice Developer

Resource Availability: Team member **Sravan** has been absent since **Day 1** of Sprint 1. As a result, **six assigned tasks remain incomplete**, creating a backlog in the planned deliverables.

Task Reallocation: To minimize schedule impact and ensure Sprint 1 completion, the pending tasks will be **transferred on Friday evening** to **Deepak, Seshadri, and Keerthan** for execution and completion.

Project Impact: The pending tasks are interconnected with multiple project modules. Delayed completion may affect dependent activities across **AI/ML development, data processing, documentation, system integration, and testing**, potentially impacting the overall sprint timeline and coordination between teams.

2.From GIS Engineer

Encountered severe system loading issues and plugin connectivity failures while attempting to extract and render large-scale vector datasets (roads and water bodies).

3. From SEO/SEM

Lack of access to premium SEO tools (such as SEMrush or Ahrefs) limited the ability to perform an accurate keyword ranking analysis.

8. Pending Tasks

1. Due to Sravan's absence since Day 1, six assigned tasks remain pending and could not be completed within Sprint 1.
2. Competitor keyword ranking analysis (awaiting access to SEMrush/Ahrefs or similar SEO tools).
3. GIS Engineer
 - Prepare base layers (administrative boundaries, roads, water bodies).
 - Create attribute tables.
 - Standardize layer naming conventions.

Encountered severe system loading issues and plugin connectivity failures while attempting to extract and render large-scale vector datasets (roads and water bodies).

9. Sprint Progress

Sprint Activity	Status
Sprint 1 Overall Completion	96.89%
Foundation Setup	✔ Completed
Dataset Preparation	✔ Completed
Research	✔ Completed
Development	In Progress

10. Conclusion

Sprint 1 of the **CARIVIX AI – AI-Powered Research, Intelligence & Decision Support Ecosystem** has been successfully completed with an overall progress of **96.89%**, demonstrating strong collaboration and consistent performance across all functional domains. The team successfully established the project's technical foundation by completing core activities related to Python development, Artificial Intelligence, Machine Learning, Data Science, GIS Engineering, NLP, Data Analytics, Research, Technical Documentation, Graphic Design, and SEO/SEM.

Most of the planned deliverables were completed on schedule, including development environment setup, dataset preparation and validation, machine learning experimentation, AI research, GIS data configuration, documentation standards, developer guides, API templates, branding assets, and content strategy. These accomplishments provide a robust foundation for the implementation phase and significantly reduce project risks in subsequent sprints.

A few activities remain in progress due to resource constraints, system limitations, and the absence of one team member. The pending NLP tasks have been planned for reassignment to ensure timely completion, while the remaining GIS and SEO activities will be addressed in the next working cycle. These issues have been identified early, mitigation plans are already in place, and their impact is being actively managed.

Overall, the successful completion of Sprint 1 reflects effective planning, strong inter-team coordination, and a high level of commitment from the project team. With the foundational infrastructure, documentation, research, and development activities now established, **CARIVIX AI is well-positioned to enter Sprint 2, focusing on feature implementation, system integration, advanced AI model development, and platform functionality while maintaining the project's planned timeline and quality objectives.**